Summer Project

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1 Background Information

1.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis (z-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{2}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{3}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{4}$$

Here w_0 is the beam waist, $w(z)$ is the beam radius at position z, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and z_R is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{5}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and A absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{6}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix T, and the detector measures only intensity:

$$E_{\text{out}} = T\,E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{7}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

1.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{8}$$

where ρ is the density, c_p is the specific heat capacity, κ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{9}$$

where n_0 is the reference refractive index at temperature T_0, and dn/dT is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{10}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{11}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

1.3 SLM Behaviour

The SLM acts only in the transverse (x, y) plane and performs phase-only modulation. The applied phase pattern is $\phi(x, y)$, so the output field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

This phase pattern is the control input to the system.

2 Experimental Setup

A setup of our system is shown below in Figure 1.

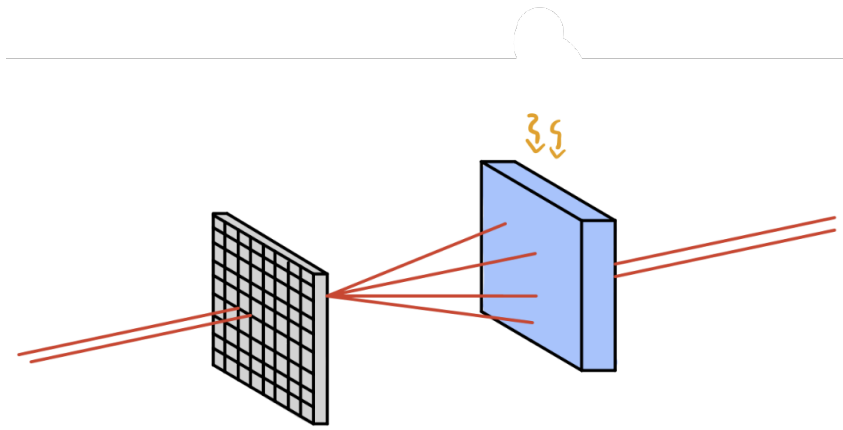

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the some material, and is detected by the camera.

The optical process is summarized as follows:

(I) Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

(II) Apply SLM phase modulation to obtain $E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}$.

(III) Propagate through the thermo-optical material using the transmission matrix $T_{\text{opt}}(t)$ to give $E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}$.

(IV) Compute the output intensity $I(x, y) = |E_{\text{out}}(x, y)|^2$.

(V) Define the reinforcement-learning reward as the intensity at a selected target output location.